Arnab Kanti Tarafder

Ph.D. Candidate in Computer Science • Distributed LLM Training Systems & HPC

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RESEARCH SUMMARY

I build systems for large-scale model training. My research develops new parallelism techniques and memory/communication optimizations for distributed Mixture-of-Experts (MoE) training and inference on heterogeneous multi-node GPU/CPU clusters. Earlier, I worked on GNN training systems and half-precision CUDA kernels (**HPDC 2025**, **USENIX ATC 2025**).

EDUCATION

William & Mary Jan 2023 – Present
Ph.D. in Computer Science (Candidate) *Williamsburg, VA*

- Passed candidacy in January 2025. Advisor: **Dr. Jie Ren**
- Summer Research Fellowship, William & Mary (2023, 2024)
- Coursework: High Performance Computing, Deep Learning, Deep Transfer Learning, Systems for DL, ML Systems

Rajshahi University of Engineering & Technology (RUET) Jan 2015 – Sep 2019
B.Sc. in Computer Science & Engineering (CGPA: 3.76 / 4.00) *Rajshahi, Bangladesh*

RESEARCH EXPERIENCE

Research Collaboration (remote), Barcelona Supercomputing Center (BSC) Sep 2025 – Present

- Collaborate with BSC researchers on memory-efficient distributed MoE training, resulting in **RelayMoE** (under review).

Graduate Research Assistant, William & Mary Jun 2024 – Present
Advisor: Dr. Jie Ren (jren03@wm.edu)

- Design new parallelism strategies for LLM/MoE training and inference that exploit heterogeneous GPU/CPU resources.
- Profile MoE bottlenecks and design memory and communication optimizations for multi-node clusters.

Graduate Researcher, William & Mary Jan 2023 – May 2024
Advisor: Dr. Pradeep Kumar (pkumar@wm.edu)

- Optimized GNN training in half precision (FP16), building custom low-precision CUDA kernels.

PUBLICATIONS

Memory-Efficient Expert Routing for Distributed MoE Training **Under review, 2026 (arXiv)**
Arnab Kanti Tarafder, Jaume Guasch-Martí, Gokcen Kestor, Jie Ren

- Proposed **RelayMoE**, a new ring-based parallelism for MoE that adaptively circulates tokens or experts across devices, removing the memory bottleneck of all-to-all dispatch.
- Outperforms state-of-the-art MoE systems (Megatron-LM, DeepEP, Tutel, FasterMoE) on 8B–57B models across **64 GPUs (8 nodes)**: **up to 2.02× throughput** and **2.85× longer trainable sequences**.

Optimization of GNN Training Through Half-precision **HPDC 2025**
Arnab Kanti Tarafder, Yidong Gong, Pradeep Kumar

- Introduced half-precision training for GNNs, achieving a **2.3× speedup** and **2.67× memory savings** over the state of the art with no loss of accuracy.

Identifying and Analyzing Pitfalls in GNN Systems **USENIX ATC 2025**
Yidong Gong, **Arnab Kanti Tarafder**, Saima Afrin, Pradeep Kumar

- Analyzed the system-design pitfalls that cause poor training performance in state-of-the-art GNN systems.

SKILLS

Languages	Python, C++, C, CUDA
Distributed Training	PyTorch, Megatron-LM, FSDP; 5D parallelism; mixed precision; communication/computation overlap; HPC clusters

TEACHING EXPERIENCE

Graduate Teaching Assistant, William & Mary Jan 2023 – May 2024
Computer Organization & Architecture; Software Development